

Behaviour & Information Technology

ISSN: 0144-929X (Print) 1362-3001 (Online) Journal homepage: www.tandfonline.com/journals/tbit20

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To cite this article: Stefan Johansson, Marika Jonsson, Jan Gulliksen & Catharina Gustavsson (04 Jun 2025): User participation in co-design – requirements for accessible online collaboration: an exploratory study, Behaviour & Information Technology, DOI: 10.1080/0144929X.2025.2511734

To link to this article: https://doi.org/10.1080/0144929X.2025.2511734

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Published online: 04 Jun 2025.

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BEHAVIOUR & INFORMATION TECHNOLOGY
https://doi.org/10.1080/0144929X.2025.2511734

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User participation in co-design – requirements for accessible online collaboration: an exploratory study

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ABSTRACT

The aim of this study was to describe prerequisites for online collaboration for user participation in co-design of online digital services as perceived by people with impairments and to develop requirements for how online collaboration should be designed to be accessible for all participants. An exploratory study was conducted in parallel to activities in two co-design projects. Researchers, accessibility experts, designers and 31 members of disability organisations participated. All participants had lived experience of impairments. Data was co-analysed by all participants using a qualitative thematic analysis with an inductive approach. Sixty-three requirements for accessible online collaboration were identified pertaining to four themes: digital tools and accessibility; preparation of an online collaboration activity; performing and participating in an online collaboration activity; and documenting and evaluating an online collaboration activity. Conclusion: the co-design process can be strengthened by transition to online collaboration, owing to the possibility of having more frequent interaction, economising the participants' energy and the design process being more transparent. Using accessible online tools, careful planning, and meeting preparation, and establishing a trustful and convivial atmosphere during each activity, are the cornerstones for accessible online collaboration. People with impairment can participate in online collaboration if the activities meet accessibility requirements.

ARTICLE HISTORY
Received 2 February 2023
Accepted 14 May 2025

KEYWORDS
Participatory design; accessibility; E-Health

1. Introduction

The two projects (CoDeAc) and Requirements for Cognitive Accessibility (DigiKog) (Begripsam 2022; CoDeAc 2021) had just started undertaking activities involving user participation with people with impairments, aiming at co-designing an eHealth application (Project A) and exploring the nature of cognitive accessibility in digital environments (Project B), when the COVID-19 pandemic prevented all possibility of collaboration by on-site activities at physical meetings. The project management in both projects had to decide between putting the project on hold or trying to shift to online collaboration. The decision was to turn to online collaboration activities. Due to impairments, the participants had diverse accessibility needs which raised questions about how to organise and perform the online activities to become accessible for everyone. This created an opportunity to investigate how to shift to online collaboration when some participants have impairments.

Within the field of Human Computer Interaction, the work in the two projects and our study was founded in the Participatory Design, PD, approach (Robertson and Simonen 2012) and framed as a combination of PD and Participatory Action Research, PAR (Balcazar et al. 1998). The specific way of carrying out the activities in the projects can be described as co-design, defined by Sanders and Stappers as 'the creativity of designers and people not trained in design working together in the design development process' (Sanders and Stappers 2008).

PD originate from the idea that people affected by the outcomes of a design should also be part of the design process to balance the power relations between different stakeholders (Bødker and Kyng 2018; Robertson and Simonen 2012). Bringing designers, experts and users together in participation could make design a living knowledge that can emancipate people to enjoy both design and designing (Lee 2008). PD has a democratic ambition, but PD has also a pragmatic stance (Bratteteig

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et al. 2017). It assumes that the quality of the design outcome will be improved if the intended users can influence the design process, making PD a strong influence for later user centred design processes.

The aim of this study was to describe prerequisites for online collaboration for user participation in co-design of online digital services as perceived by people with impairments, and to develop requirements for how online collaboration should be designed to be accessible for all participants.

2. Background

User participation in co-design processes traditionally relies on physical meetings, and although emphasising creativity and novelty, the design process itself can be described as conservative and following similar paths every time (Steen, Manschot, and De Koning 2011). Workshops are seen as fundamental for any design process (Klammer, Van den Anker, and Janneck 2011) and a team of people drive the process further towards the intended design by a consecutive series of workshops. Even with high costs for travelling and accommodation, the idea of having workshops to gather people in the same place has not really been challenged. As user participation and user centred approaches have gradually gained increased importance in any design process (Kujala 2003; Kushniruk and Nøhr 2016; Sanders and Stappers 2014), users have been invited to physical workshops when contributing to co-design activities. That was until the COVID-19 pandemic shut down society and restrictions made physical meetings impossible or inappropriate.

2.1. Distributed participatory design

PD in the making is almost always situated to a specific time and a specific place, but there is a growing interest in Distributed Participatory Design (DPD) (Bergvall-Kåreborn and Ståhlbrost 2008; Danielsson et al. 2008; Slingerland et al. 2022; Walsh 2011). An early example of DPD was design probes or cultural probes, as a potential alternative for data collection when participants were unable to meet physically (Gaver, Dunne, and Pacenti 1999; Sanders and Stappers 2014). Probes allowed people to create and record ideas and thoughts from home using a wide range of materials and tools. The ideas could then be sent to the researchers as part of the data collection process.

Klammer et al. used DPD to move away from singular face-to-face events and into an action research setting by using digital tools to support continuous collaboration in service innovation (Klammer, Van den Anker, and Janneck 2011). They applied a blended co-design process with a combination of face-to-face and virtual methods, illustrating that a development of DPD methods become possible with the introduction of online tools for meetings and online tools for collaboration.

The COVID-19 pandemic pushed the need for DPD. Due to restrictions in meeting possibilities, collaborative design projects either had to be postponed or turned to online activities (Alves-Oliveira, Wiesmann, and Mihata napoli, italy. 2022; van Nifterik, Visser, and van Erp 2021). So far, there is only a small body of research concerning online participatory approaches during the COVID-19 pandemic.

To do PD during lockdowns and restrictions, Constantin et al. (2021) explored DPD, with participants in the design team that was geographically dispersed and presented how they shifted to an online workshop. They found this to be an opportunity to address challenges caused by the pandemic. They summarise seven key themes of opportunities and challenges in DPD: Participation in online environments; Maintaining engagement; Sense of connectedness/togetherness; Accessibility, diversity and inclusion; Power dynamics; Developing skills; and Administration, pragmatics, and logistics.

A narrative review from 2021 by Hall, Gaved, and Sargent (2021) found 38 studies and 4 of those presented research projects that shifted from physical to online activities. Al Masri and Masannat (2020) described how they shifted format for data collection from in-person interviews to online interviews using an adaptation of an online education platform. Dodds and Hess (2020) presented how they shifted to online group discussions with vulnerable people and their support groups. They suggested that online group interviews are potentially comfortable, non-intrusive, safe, engaging and convenient if technology works. But they also highlighted limitations, such as lack of non-verbal communication, privacy issues and access issues. Nicol et al. (2020) recommended that researchers implement physical but not social distancing and keep the research activities going by shifting to digital tools. They argued that minor protocol deviations and disruptions can be handled by model flexibility. Woodward, Harrison, and Tope (2020) presented a case from Zimbabwe that used a text-only WhatsApp group to have a 2-hour workshop and highlighted how the work was facilitated because of low bandwidth and affordability.

In the review study, Hall et al. found no reports of going online to co-design together with people with impairments, but they found literature presenting more general challenges in online participatory activities during the pandemic (Hall, Gaved, and Sargent 2021). Participants perceived participation as both

stressing and as an opportunity to have something to do during lockdown (Lupton 2020). Barriers to online activities were participants having low information technology literacy (Al Masri and Masannat 2020; Marzi 2020; Warner-Mackintosh 2020; Woodward, Harrison, and Tope 2020) and that impairment could cause problems with accessibility (Woodward, Harrison, and Tope 2020). Hall, Gaved, and Sargent (2021) suggested that a solution could be to arrange separate sessions to train participants prior to online activities or to produce tutorials to help participants with the skills needed for online participation.

Slingerland et al. used an online summer school for Irish teenagers making artwork to reflect on challenges and opportunities for DPD (Slingerland et al. 2022). They found that DPD can be inclusive, create flexible opportunities for participation and for the use of digital tools. For this to happen, Slingerland et al. argued that facilitators need to be careful in planning and performing online activities, and in building trust and creating relationships. They called for more work on how to organise distributed workshops (Slingerland et al. 2022).

2.2. Physical meetings are the default option for co-design activities

In the literature, user participation, collaboration, focus group discussions and workshops are almost always described as activities taking place in a physical location. Krueger and Casey (2009) describe an appropriate environment as something including both physical and socio-political environments. A comprehensive tutorial about focus group methodology does not mention the possibility of them being conducted online (Liamputtong 2015). Wilson et al. present a number of co-design techniques to include people with communication impairments (Wilson et al. 2015). For all the techniques presented, though it is not explicitly outlined, the reader understands by looking at pictures included in the article that all the workshops described were physical meetings. It seems so obvious that research presenting collaboration activities seldom mention the fact that people were gathered in the same physical room to participate in the activity (Krueger and Casey 2009; Kushniruk and Nøhr 2016; Liamputtong 2015; Steen, Manschot, and De Koning 2011; Wilson et al. 2015).

2.3. Scaffolding and preparation to create accessible meetings

People with impairments might have special requirements for user participation in a context of co-design. According to Johansson (2019), careful scaffolding and preparations can create accessible meetings and enable people with impairments to take part in co-design activities. Such activities are typically carried out by gathering the participants in the same room, using whiteboards, sticky notes, pens, papers, projections on a wall and sometimes the division of participants into small groups to brainstorm and elaborate on ideas. Johansson found that the structure of activities needs to be carefully planned and executed. Also travel arrangements to and from activities, pauses during activities, and food need careful attention. Otherwise, many participants, especially those with cognitive impairments, could not participate or had to quit participation halfway through an activity.

Thus it is reasonable to believe that some people with impairments would find it difficult to shift from physical to online participatory activities, and also that online collaboration activities need scaffolding and preparation that are specific to the online setting. How preparation and scaffolding of online participatory co-design activities should be carried out is explored in this study.

3. Methods

3.1. Study design

The study had an exploratory study design (Stebbins 2001), using PAR methods (Macdonald 2012). PAR is characterised by its democratic approach, and it allows for people with impairments to take an active role in defining the important problems that need to be solved. PAR comprises an accurate and authentic co-analysis of the problem and problem solutions in a way that improves the quality of life for the people and communities taking part (Balcazar et al. 1998). The study drew from a theoretical critical realism approach focusing on the mechanisms causing disability, but also acknowledged that reality is influenced by cultural and social expectations and interpretations (Bhaskar and Danermark 2006). Within PAR and PD, we used a process where co-design methods were moved to online collaboration to be able to discuss the requirements for participation in this type of DPD.

3.2. Participants

The participants in this study were part of two co-design teams in the projects A and B,CoDeAc and DigiKog. As a consequence of the COVID-19 pandemic both projects had to shift from physical to online collaboration. After initial testing, the projects used Zoom (n.d.) and Miro (2021) as tools for online meeting and online collaboration.

The aim of project A was to co-design an eHealth application that would propose a better design of the Swedish eHealth portal 1177.se (2023). Participants with impairments formed a design team together with a designer, an accessibility expert and a researcher in human computer interaction and co-designed a hi-fi prototype. The prototype was developed in a series of online co-design workshops and was used as a benchmark against today's solution. Four researchers, two accessibility experts, two designers, and 31 members of two disability organisations participated in the study and formed two working groups. The age span of the participants was 22–73 years, and all participants had lived experience of impairments. Meetings were led by a facilitator. The facilitator was assigned to the role before the meeting. The researcher and the accessibility expert acted as facilitators. In meetings using breakout rooms for work in smaller groups, the facilitator role could also be assigned to a participant with previous experience of leading discussions.

The aim of project B was to apply the ISO standard for cognitive accessibility (ISO 2020) on public online services in Sweden. Series of three online co-design workshops was arranged in collaboration with government agencies, regions and local municipalities to propose re-design or a new design of e-services. In project B, a researcher or the project leader acted as facilitator. The project also used a documenter role. This person took screenshots and made notes during the discussions.

A broad range of impairments related to cognition, perception and mobility were represented among the participants in both projects. Many of the participants had multiple impairments. Those impairments could potentially create problems with using technology, communication and interaction with other people, or to understand information and instructions. Hence, the accessibility and usability of a co-design activity need to fulfil high demands to be perceived acceptable. Most of the participants used a computer and connected from their home when participating in online activities. Some participants used a smart phone or a tablet and connected from home and occasionally also from other places such as outdoors or on a train or a bus.

The participants in both projects knew each other before the start of the online collaboration. They had previously met on physical workshops. The participants belonged to a community engaged in the promotion of better design of online services to make the digital society more inclusive. They had expressed an interest in participating in either project A or B and some of the participants had also been part in the decisions and planning and the application for funding of the project before they started. Participation in the workshops was offered payment, according to principles established in collaboration with the disability movement, public sector bodies and the MedTech industry (Johansson et al. 2022).

All participants gave written informed consent before taking part in the study. The study was approved by the Swedish Ethical Review Board (ID-no: 2019-04419).

3.3. Procedure, data collection and data analysis

The study was conducted as a parallel process in collaboration with two co-design projects. All project activities, mostly workshops and group meetings, shifted to online collaboration in the spring of 2020. Participants used a variety of hardware; computers (Mac and PC), tablets (Samsung and iPad) and smartphones (iPhone and Android) to participate in the online meetings. The number of participants in each online session varied from 4to 5 in some sessions and up to 15 in others.

As a parallel process, participants discussed and analysed the experiences of shifting to online collaboration, and how online co-design activities should be carried out to be perceived as accessible for all participants. The work was gradually summarised as experiences of benefits for online collaboration, and to the development of a checklist on requirements for online collaboration. At the end of each online workshop in the two projects, reflections on the process of the online collaboration were expressed and documented. All participants got the opportunity to share their reflections orally and the facilitators took notes for documentation. Participants could also send reflections after the workshop or book a meeting with the facilitator to share their reflections. The facilitator used questions to get the reflection going: 'What could have been made in a better was at today's session?' and 'What was good about today's session?' A Miro-board (Miro 2021) was used to present the collected data. All participants had a link to the board and could access and comment on it at any time. After each analysis session, the facilitators updated the board with new information. In addition, two specific workshops were arranged where the participants discussed experiences of online collaboration. A final workshop was conducted to find consensus and agree on the final set of requirements. To avoid hierarchies, all participants were regarded as co-researchers having equal voice at analysis sessions. We frequently returned to issues to reflect and discuss until consensus was reached.

The co-analysis of data was conducted in an iterative process using a Miro-board to identify patterns, and reflect on codes and patterns, and followed a qualitative thematic analysis with an inductive approach (Braun

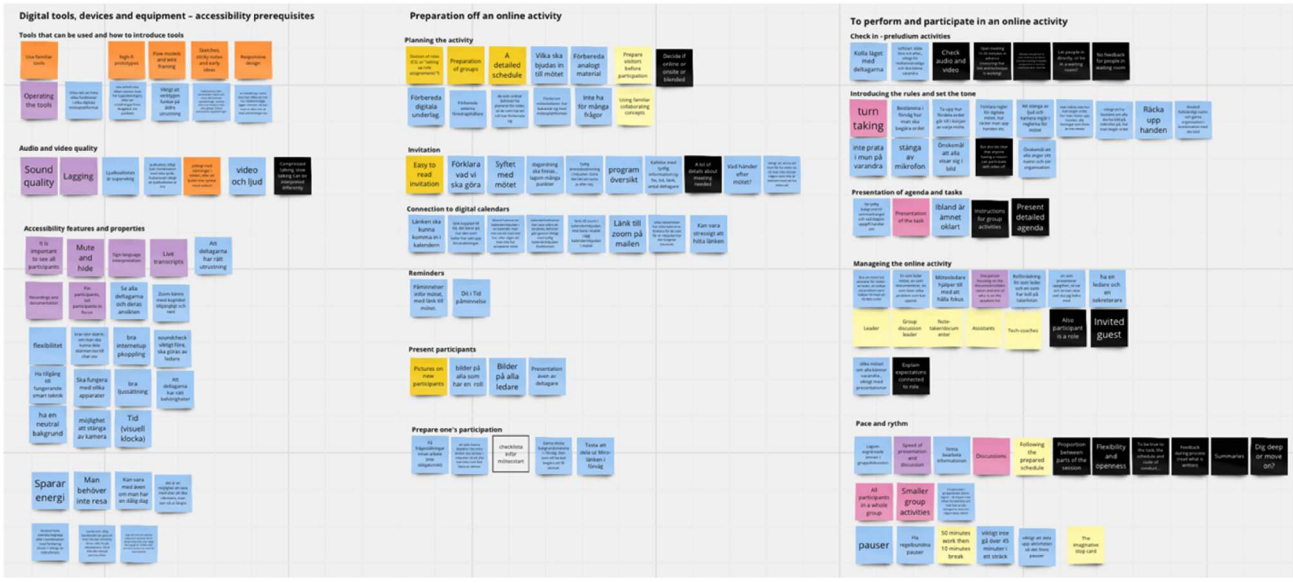


Figure 1. Visualisation of the thematic co-analysis process by a screenshot from the Miro Board.

and Clarke 2006). Two separate thematic analyses were undertaken relating to each of the two objectives: one analysis of perceived benefits of online collaboration and one analysis of requirements for online collaboration. In a first step, participants discussed the data and started to create codes from the data on a semantic level. In the next step, codes were interpreted and organised into meaningful patterns by the participants. The patterns were named as themes and iteratively revised until consensus was obtained. Figure 1 shows an example of the thematic co-analysis process by a screenshot from the Miro Board. In the Miro board, all participants could ask the facilitator to move around sticky notes that outlined the codes, or to change the name of a theme or a subtheme, or to add new themes.

4. Results

The result section is structured in the following way: first we report on how the participants described the benefits of online collaboration. Then we describe the requirements for accessible online collaboration. Those requirements are divided in 4 themes, comprising 14 categories and 63 requirements. The result section ends with a presentation of the most important requirements.

4.1. Experiences of benefits for online collaboration

The participants identified a number of benefits of online collaboration when shifting from physical meetings to online meetings.

Economising on energy was the most prominent benefit. Many expressed that having to travel to a physical meeting consumed a lot of energy that could instead be allocated to the tasks in the online activity. Participating from home also created a feeling of safety and familiarity and the absence of unwanted surprises that can occur when travelling to a meeting or trying to locate the facility for the meeting. It also created more control over breaks and meals. One participant expressed how online collaboration opened new possibilities for flexible participation when participating from home:

> Online is better, I can participate even when I'm having a bad day. (P4, during an analysis session).

Problems with fatigue were more easily handled in online activities than in physical meetings. Participants that ran out of energy could disconnect and reconnect later or disable their video and lay down and listen to the workshop discussions.

People living in different parts of the country could easily form a group and work together without having a budget for travel and accommodation. The only time needed for online collaboration was the time spent for the participation in the workshop, as compared to travelling to a physical location which could result in a 1- or 2-day absence from home.

Another benefit was the ability to have several shorter meetings with time to reflect, or for developing prototypes or sketches between meetings. Having for example one co-design workshop a week is possible when working online but often outside the possible scope when having physical meetings. Many participants enjoyed the possibility of using built-in non-verbal feedback

features in online meetings, such as thumbs up, a heart or clapping hands or raising a hand.

Having material on a shared space online provided the possibility of interacting with the material at any time. Several participants noted that they could get ideas and think about important issues several hours or days after a meeting and that the online space provided an opportunity to document this at any time, also after the meeting.

On a personal level, many participants noted that they improved their digital skills. They became more confident in using the digital tools and solving technical problems. Some of the participants also used their new skills to teach and support less experienced peers. They also used digital tools in other situations. Several participants noted an improved self-confidence in using digital tools and participating in online meetings.

Many of the participants called for more online meetings outside the project work. Some of them initiated digital coffee-break sessions, meetings just to hang out as a means to break social isolation during the COVID-19 pandemic. The urge for more social online interaction was also manifested by participants starting to use online meeting tools for example to give each other walk-a-rounds in their homes and present paintings they were working on.

4.2. Requirements for online collaboration

Based on the experiences from the participants, 63 requirements for accessible online collaboration for user participation in co-design were identified. The requirements are organised in four themes and are presented in a chronological order beginning with device preparation, followed by the implementation and evaluation of co-design. The four themes are: digital tools and accessibility; preparation of an online collaboration activity; performing and participating in an online collaboration activity; and documenting and evaluating an online collaboration activity (Table 1). The requirements can be applied in a single online design session as well as series of design workshops during the whole cycle of a design process.

4.2.1. Theme 1: digital tools and accessibility

Several barriers for online collaboration were identified. Most of the barriers were related to the digital tools that were used for online collaboration (that is Zoom and Miro) and occurred mostly during an introductory phase. When people became familiar with the tools and if the tools did not change, the collaboration could run smoothly. Some participants needed to be introduced to the online tools in separate training activities, before the actual participation in the online activity. A commonly occurring barrier was that some participants perceived that other participants were talking too fast.

4.2.1.1. The participants' devices and internet access. Participants used their devices and internet connection and most often they accessed the meetings from their home. Some had relatively old equipment, which could pose problems, since some digital tools mandate that users have modern equipment. Participation was dependant on a good internet connection and most of the time the quality was considered good enough, even if almost every meeting had some participant struggling with lagging sound. The participants found it to be easier to access to the meeting from a computer compared to a tablet or a phone.

4.2.1.2. Functionality in the digital tools. Understanding the procedure for how to download a digital tool or updating to a new version of a tool was not self-evident, and some participants needed guidance and support to do this. New problems could emerge every time a tool was updated with a new version. New functions took time to introduce and changes in a familiar interface took time to re-learn. It could be confusing when participants were using different versions of the online collaboration tools.

> It is not easy to know which version of a tool I'm using, if it needs to be updated, and how to access different adaptation possibilities. (P17, during a design workshop)

The managers of the online activity have to master the tools and be familiar with how to enable accessibility features to pass that knowledge to participants. An aspect that was discussed among the participants was that some might need to adapt the online collaboration tools to fit their personal needs and that people using Assistive Technology (AT) might need help to adapt tools and AT to work together. A specific problem was that many digital tools for collaboration do not have a Swedish language version. Some of the participants found English language versions very difficult to use.

Among non-verbal feedback functions, the 'raise hand' function was the most important for running a meeting smoothly. It was used for permission to speak requests and helped to create an order of speaking for the participants to follow, and for the meeting manager when distributing the word among the participants. It was very important to have one person speaking at a time. As participants got more familiar with online

Table 1. Requirements for accessible online collaboration for user participation in co-design.

Main theme	Category	Requirement
Digital tools and accessibility	Participants' devices and internet access	Participants must have devices able to perform online activities
		Participants must have a fast enough internet connection
	Functionality in the digital tools	Use digital tools that are perceived usable by the users with impairments*
		If possible, use tools with non-verbal feedback interaction features
		All participants should be able to be displayed in a video frame or picture
		If possible, use tools that have native language versions
		Ensure that participants have permissions to use functions in the tool
	Accessibility features in the digital tools	Evaluate the accessibility in new features before starting to use them*
		Live transcripts should be easily available
		A deaf person and a sign language interpreter should be able to set each other in focus, alongside with current speaker
		Screen reader accessibility should be available
		Keyboard navigation should be available
	Audio and video quality	Provide an audio quality that makes it possible to clearly hear online conversations*
		Provide a resolution for the display of text and visual details that are big enough for participants to see. Describe what is displayed for those who cannot see
		Ensure that meeting managers do sound check with all participants before start
		Ensure that participants and meeting managers can mute microphones
		Participants should have a neutral or blurred background
		Ensure that participants have enough light around them to be properly displayed
	Help and support	Allow participants to test the tools that will be used in advance*
		Help participants to adapt the tools according to personal preferences
		Help participants to download and install new versions of the tools
		Help participants to understand new features or changed functionality in the tools
		Give participants access to tutorials in text and video in easy-to-understand language
		Dedicate a specific support person in online collaboration activities*
		Allow participants to communicate with a support person between online collaboration activities to solve problems prior next activity
		Provide a support channel outside the digital tool during ongoing online collaboration activities (e.g. phone, SMS chat, etc.)
Preparation of an online collaboration activity	Planning the activity	Prepare an agenda for the activity where available time and tasks are balanced (e.g. subjects, times, pauses, guests, roles and people needed to manage the activity)*
		Cast the activity with proper roles to manage the meeting
		Consider the need of interpreters or transcribers
		Prepare material that will be used in the activity and provide it in accessible formats
		Plan for how to document the activity and how to receive feedback
		Prepare guests and new participants for taking part
	Invitation to the activity	Produce a detailed but easy-to-read invitation
		Present all participants with images and short bio
		If needed, send instructions and material for participants in advance
		Send invitations in different modalities adapted to the participants accessibility needs
		Use reminders to help participants attend the activity
Performing an online collaboration activity	Check in and check out	Open the meeting some 15–30 minutes before the formal starting time
		Use direct access to a meeting rather than a waiting room
		Create a soft and welcoming start to set a convivial tone*
		End an activity by summarising what has been done and tell what will happen next
	Introducing the rules and set the tone	Present the agenda and the upcoming tasks, introduce guests, etc.
		Present how the activity is supposed to work
		Present potential accessibility enhancements
	Pace and rhythm	Follow the prepared schedule, end in time or declare why the plan must be changed
		Be flexible and open to adjustments in plans if reality show they doesn't really work
		Adjust the speed of presentations to allow all participants to grasp what is going on
		Consider when to move on and when to dig deeper on a topic
		Allow everyone to speak, but respect if someone does not want to speak
		Allow people time to reflect and formulate what they are thinking
		Regularly summarise work so far, repeat important information when needed
		Allow every participant the right to stop an activity to discuss formal procedural issues
		Allow every participant to contribute in a way that releases their full potential
		Consider a mix of working in the whole group and smaller groups
	Collaboration strategies	If possible, do live changes of the material you are working with
		If possible, use online collaborating tool that can share material with all participants
		Take benefit of the online collaborating tools possibility for asynchronous work (e.g. let participants use a shared space before, during and after the activity)
		Document by using accessible formats allowing participants to reflect between activities and take part in analysing the material
		If possible, use a rich spread of documentation materials (e.g. text, images, videos, wireframes and prototypes)
Documenting and evaluating an online collaboration activity	Documentation	If possible, use the online collaborating tools potential of instant documentation (e.g. digital post-it notes, sketches, uploaded material, recordings, etc.)
	Reflection	Facilitate for reflection after an activity and after the final work is done
	Evaluation	Evaluate and reflect on how the online collaboration works on a regular basis

*Requirements which were considered the most important by the participants.

meetings, some started to use other non-verbal feedback such as 'thumbs up' and 'applause'. Most participants appreciated the 'raise hand' function as a request for permission to speak, but some struggled to remember how it worked and that it existed. This was solved by a recommendation to use the function; however, it was permitted to physically raise a hand, or write in the chat, or just say their name to notify their wish to be on the speaker's list.

It was important that the online meeting tool could display all participants' videos and have functions for screen sharing The host should be able to mute and unmute the participant's audio or video. Any participant should be able to mute and unmute themselves. Anyone in the meeting should be able to share content from their device. The presenter should be able to enlarge or zoom in on content, if requested by participants. It was also important to be able to see both the displayed content and the participants at the same time, but this was problematic for participants using tablets or phones who had to toggle between content and participants.

There was mixed opinion among the participants about chat functions. Some regarded ongoing chat activities as impeding the main process, since it distracted attention from the main activity and consumed a lot of energy, while others liked the possibility to get explanations, further comments, links, etc. in the chat. Even the small red object, notifying new posts in the chat, was perceived as stressful by some participants. This was solved by allowing participants to request that chats should not be used. And if such a request was made, the meeting manager encouraged the rest of the participants to respect that request.

When participants addressed conflicting needs, as in the example with the chat, the agenda was set on pause and the issue was discussed. An ambition then was to try to reach a consensus on how to handle the situation. As a guiding principle it was decided that if something would totally prevent or significantly restrict someone from participation that something should be avoided. As for the example with chat, those who liked using the chat could accept not to use it when they understood that it was a major problem for others. In another example, when there was a discussion about speaking slowly the solution was to recommend participants to talk in their natural way but try to make a bit longer pauses between sentences.

For prototyping and other design activities, the designer's preferred choice of tool was used. Our lead designer preferred Adobe XD but we also worked with guest designers using Figma and Axure. Instead of every participant having to learn a design tool, participants could remotely ask the designer to make changes and additions in the design. More extensive design work was done between online meetings by the lead designer or guest designers, and then presented to the rest of the participants at the next online meeting. After a while, the participants started to do simple wire framing or upload hand-drawn sketches in Miro. It was also found that Miro could be used as a collaborative space and a space for collecting data such as sketches, design proposals, pictures, sticky notes, etc. This allowed for both synchronous and asynchronous use of the space. Since all participants received the link to the Miro board, they could visit the shared space at any time without having to handle login credentials.

4.2.1.3. Accessibility features in the digital tools. To use online meeting tools that met advanced accessibility requirements, the participants in the project groups evaluated several tools (Microsoft Teams, Google Meet, WebEx, Mural) before deciding to use Zoom for online meetings and Miro for online collaboration activities, sharing of material and co-designing. These online tools were perceived the most accessible by the participants. In particular, the cognitive accessibility was considered the best in these tools.

At some meetings, there were participants that required live transcripts of the discussions. For example, when someone used a difficult dialect or had difficulties to hear the dialogue. Technology for automated live transcripts was tested but they were not perceived good enough by the participants, at least when transcribing the Swedish language to text. Hence there was a need to add human live transcribers to some meetings. At some meetings, sign language interpreters were needed, and the deaf participants and the interpreters found it important to be able to lock or pin each other so that they could always be in each other's focus.

Blind participants using screen reader technology required that the tools should be able to operate with keyboard access and that worked reasonably well in Zoom. In many cases, there was a need for additional live audio descriptions since the content discussed was often of a visual nature. This was the case when people with visual impairments took part in discussing content on a Miro-board.

4.2.1.4. Audio and video quality. The audio quality was more important than the video quality. Lag in discussions or poor sound quality had the potential to ruin an online activity.

> It is super-important that the sound is working. If something that someone says gets lost during

> transmission it can be really confusing and annoying. (P12, during an analysis session)

A headset provided for the best audio quality and also prevented disturbing background noise. Disturbing visual backgrounds were identified by the participants as something that could potentially draw attention from the person in focus. Blurring a 'noisy' background or using a neutral background image was found useful to prevent disturbance. Good lightning conditions were important to see the participant's face, notice shifts in facial expressions, and, if needed, lip read. It was recommended to mute when not speaking.

4.2.1.5. Help and support. There was a learning curve when participants started to use the online tools and a need for remote support for the participants was found important. A specific role for assistance and support was introduced, called the Tech Coach (TC). The TC was experienced in using technology and in how to explain to others how to use technology. Participants could reach out to the TC at any time. It was important that the TC could get in touch with the participants outside the meeting tool if they for example struggled to connect to an online meeting. Therefore, the TC had a list with participants' telephone numbers, email-addresses, and contact data to their social media applications.

> It is useful to have the telephone number to the TC, so I can call if for example I can't log in to a meeting. (P7, during an analysis session)

It was regarded as important that the TC understood the diverse needs represented in the group of participants and could adapt to different levels of maturity in using Information and Communication Technology. Alongside the TC support, participants engaged in helping each other to master the different digital tools. The TC created easy-to-read tutorials for the digital tools, complemented with images. Online meetings with more than 20 participants or several new participants were staffed by two TCs.

The TC noted that different devices brought about a slightly different appearance and performance from the digital tools that were used. The TC also noted that different web browsers could cause small differences in appearance, making remote support harder. Headsets and microphone settings caused trouble in the beginning. The TC arranged preparatory training sessions, personal meetings, and online group meetings for participants to socialise and to become familiar with the technology.

A recurrent problem was to find the link to the online meeting. To maximise the ability to find the link, a strategy was developed to send meeting links in calendar invitations and e-mails, and in reminders by SMS messages. But the meeting managers were also prepared to send links to participants on request just before the start of a meeting. Another recurring problem was users struggling with turning their audio or camera on or off. The most effective solution was to ask the participant to leave the meeting and then re-join.

4.2.2. Theme 2: preparation of an online collaboration activity

4.2.2.1. Planning the activity. Careful and detailed planning of the online collaboration activity was a key for success. Online material should be prepared prior to the activity and depending on the task, participants could get a link to the material in advance. The meeting managers should consider if there was a need for additional accessible formats of the material. The meeting managers should also consider how many people would be needed to manage an activity. For example, if the planned activity involved small group discussions, more people were needed to lead the small group activities and to document activities. Also, if several of the participants took part in an online collaboration activity for the first time, more time for technical support and checking of equipment were needed. It was found that the staffing of a meeting could require for example a leader, group activity leaders, note takers, tech-coaches, and sometimes interpreters and live transcribers.

If interpreters or transcribers were needed it was important to check that they were familiar with participating in online activities. The interpreters and transcribers found it important to get some material in advance to allow them to prepare for their work. Guests and new participants should be informed in advance on accessibility issues and informed about the rules for the activity. For example, a guest that was supposed to make a presentation should be requested to create an accessible presentation and to speak slowly.

4.2.2.2. Invitation to the activity. The participants emphasised that a detailed but easy-to-read invitation should be used to present the aim of the activity, a detailed time schedule with frequent breaks, a description of upcoming tasks and instructions on how to prepare. If needed material should be sent to participants before the meeting, allowing them to prepare their participation in advance. It was important that the meeting invitation presented all participants with a photo and a short bio, especially guests and new participants.

> The purpose of the meeting should be clearly described, so that it will be easy for a person to decide on whether to participate or not. (P30 during an analysis session).

Meeting invitations had to be sent as both email and calendar invitations. In addition, we found that it was helpful for the participants to also get invitations on closed social media groups and reminders as SMS texts. A calendar invitation allows participants to easily enter the activity into their digital calendar and makes it possible for participants to set reminders of their choice in their own devices.

4.2.3. Theme 3: performing and participating in an online collaboration activity

4.2.3.1. Check in and check out. A soft and welcoming start was described as important by the participants and a way to achieve that was to open the online meeting 15–30 minutes in advance, allowing participants to test their equipment, get some support and just hang around. It was better to give participants direct access to the meeting rather than to leave them in a waiting room. Being in a waiting room cut off any possibility to communicate with anyone in the meeting management and that could create uncertainty, anxiety, and frustration. The last 5–10 minutes of a meeting should be allocated to summarising the performed activities and informing the participants of the plans for what will happen after the meeting.

4.2.3.2. Introducing the rules and setting the tone. When the online activity started, it was important for the leader of the activity to set a convivial tone but at the same time establish the rules for the activity, introduce the agenda and the tasks ahead. The term convivial refers to creating an atmosphere of companionship, trustfulness, openness to others and respectful acceptance of co-participants. If there were transcribers and sign language interpreters present, they should be welcomed and presented. Useful accessibility features in the tools should be presented, and participants should be requested to produce audio descriptions if they intend to present visual content.

It was also considered important to present rules on how to manage the discussions, e.g. if the chat should be used or not, how to name oneself in the video frame, and how to ask for permission to speak. As a general rule, everyone that was present should have their camera on all the time, and the microphone unmuted only when they had the floor. But for example, if someone was having a bad day it is was acceptable to participate without a camera on. During discussions and especially when simultaneously sharing content, someone other than the leader of the meeting needs to keep an eye on participants asking for permission to speak. If the meeting was to be recorded, the participants found it important to be notified and give their consent in advance.

> For me, it is really important that I can see the name of everyone since I do not remember names easily. If someone's name is Bxd17 it makes it hard for me and I feel ashamed for not remembering names. (P4 during a design workshop)

4.2.3.3. Pace and rhythm. The overall pace and rhythm of the online meeting was a very important aspect for how the planned activities and online collaboration played out in real time. It was important that the leader of the meeting had a feeling for when to dig deep into a topic and when to move on to the next part of the agenda. It was also important to moderate the speed of for example presentations, allowing all participants to grasp what was going on.

Pace and rhythm also concerned allowing participants time to understand the tasks and absorb given information, decide if something needed to be repeated and when to move on and when to give time for reflection. It also concerned deciding if suggested changes in presented material (e.g. a prototype) should be made instantly or if information should be collected for later changes. An important part of creating a good pace and rhythm was the combination of working in smaller groups and discussions with all participants in one large group. It was important that the online meeting tool allowed for smooth division of participants into several smaller groups, which could easily be called back into one large group. Small groups of 4–6 people were considered favourable for everyone to take part and allow everyone room for sharing their ideas.

It was very important to follow the planned agenda and the time schedule. Some 40–50 minutes of work should be followed by 10–20 minutes break to allow participants to recover energy to continue the work. Online meetings longer than 150 minutes needed a longer break allowing participants to eat and drink.

It was important to be flexible and open to adjustments when original plans did not work as intended. Such changes should be clearly announced and discussed by the participants, allowing them to create a mental re-orientation of the agenda, the time schedule, and the tasks ahead. It was preferable to plan another meeting rather than to try to either squeeze in what had turned out to be too extensive a task for the current meeting, or to prolong a current meeting.

Participants found that the leader of the meeting should be mandated to request someone who is becoming to dominate in the meeting to take a step back to allow others more space.

> It's OK for the leader of the activity to stop me if I'm taking up too much space, I know I can do that when I get engaged in something. (P11 during an analysis session)

Some participants needed a long time to think and formulate what they wanted to say. They could be quiet for a long time before they entered the discussions. Therefore, it was important to respect someone who wanted to be silent during the meeting, but at the same time some silent participants would become more active if they were encouraged to share their thoughts.

4.2.3.4. Collaboration strategies. An online collaboration meeting should not start with a blank sheet. Starting blank could be too abstract for many participants. It was considered better to start a co-design activity for example by presenting something that had been prepared in advance. That could be some early sketching or some previously existing design that could be examined. Many participants found it easier to have something to react upon, and from that reaction they could start formulating new ideas, propose solutions, and create sketches, wireframes or prototypes.

The participants found that having the designer share the design tool inside the online meeting tool, created an opportunity for all participants to propose changes in design and content that could be instantly addressed by the designer. Likewise, instant changes could be made in the production of documents for example. This was found to be better than the traditional design process in which the designer collect data during one session and returned at the next session to present the result by the way of a revised design. Direct and real-time changes in the design allowed everyone to propose and discuss changes as they took form. Also, early discussions about ideas could be conducted together by the use of a shared online space. The participants found digital sticky notes very useful for brainstorming or for analysing material, since they could be created and moved around by all participants as the discussion progressed.

By also sharing online spaces between a consecutive series of meetings, it became possible for participants to reflect, propose changes, and add comments when they wanted. Participants also found that by knowing that there would be another meeting they, became more relaxed and felt that they did not have to stress to express their ideas on one occasion.

4.2.4. Theme 4: documenting and evaluating an online collaboration activity

A useful strategy was to have note takers documenting the discussion, for example with digital sticky notes, instead of every participant taking their notes. This allowed the participants to focus on participating and still be reassured that what they were saying was being documented.

By recording online meetings and using digital sticky notes and other material on a shared digital space, the activities became instantly documented both in text and audio-visual form which could be shared by all participants. Later, a richer text document could provide the final documentation alongside the digital artefacts produced, e.g. wireframes or prototypes. It was important that all participants should be able to react on documentation or to provide reflections and ideas emerging after the online activities were finished.

Having all material on a shared digital space also made it possible for all participants to collaborate on the analysis of collected data and the interpretation of the results. This created an opportunity for all participants to also work together in data analysis.

It was important to allocate time to discuss how the online collaboration worked, and if needed, to adjust the arrangements. This could for example be done at the end of each meeting or by survey questions after the meeting. Participants also found it favourable to have specific occasions at which the only objective was to discuss and analyse how the online meetings had worked out and what could be improved.

4.2.5. Most important requirements

The participants discussed if it was possible to rank or prioritise among what now is a comprehensive set of requirements. They regard the complete set as a way to scaffold for participation, meaning that every requirement is important. Still, it is possible to put emphasise on some very important parts. In general participants found that what is presented under the section Pace and rhythm is very important. Among other requirements they regarded the following as extra critical:

- Use digital tools that are perceived usable by the users with impairments (theme 1)
- Evaluate the accessibility in new features before starting to use them (theme 1)
- Provide an audio quality that makes it possible to clearly hear online conversations (theme 1)
- Allow participants to test the tools that will be used in advance (theme 1)
- Dedicate a specific support person in online collaboration activities (theme 1)
- Prepare an agenda for the activity where available time and tasks are balanced (e.g. subjects, times, pauses, guests, roles and people needed to manage the activity) (theme 2)
- Create a soft and welcoming start to set a convivial tone (theme 3).

5. Discussion

Our study adds to the current knowledge by providing a detailed set of requirements that are relevant for inclusive user participation of all types of participants in online collaboration. We have generated this knowledge by having arranged and fine-tuned online co-design collaboration activities together with participants with a diverse set of impairments.

What started as a crisis management effort to be able to continue user participation in two co-design projects, is now the primary setup for our co-design activities, even when there are no restrictions on travelling and meeting. Turning to online collaboration made it possible to arrange more activities and workshops since the cost for the online meetings was lower than the physical meetings, and they were also less time consuming. Thereby, online meetings could be a more sustainable form of collaboration. However, in line with findings by Slingerland et al. (2022), careful preparation of the online activities is needed and we have found that new roles are needed to facilitate the meetings, for example a Tech Coach.

It can no longer be presumed that 'a workshop' or 'a meeting' is something exclusively undertaken in a physical space. For clarity we should add labelling of workshops and project meetings as online or physical, with a third option being hybrid. We can similarly not presume that materials used in design workshops are physical artefacts, they might also be in digital forms. When Sanders and Stappers compared approaches in co-design they described 'the making' as central in any approach (Sanders and Stappers 2014). It was implicit that 'the making' was going on in a physical space. This was also the case when Kushniruk and Nøhr presented rationales for user participation (Kushniruk and Nøhr 2016). Collaboration in online activities as a form of Distributed Participatory Design offers new possibilities. In our study, we found that online 'making' is indeed possible and could add qualities to the co-design process. Our work aligns with the work by Constantin et al. (2021) who found that technology and remote participation have an inclusive potential and can remove barriers for participation. We provide a holistic overview of the requirements that needs to be considered when doing online collaboration together with people with disabilities. Many of those requirements are relevant for everyone and that is expected. The theoretical model behind the concept of universal design (Persson et al. 2015) explains why an environment that is regarded as accessible for people with the widest range of need also will benefit all. Nevertheless, it is possible to distinguish some of the requirements as extra important when participants are having disabilities. Most of the extra important requirements can be related to functionality and accessibility, but the participants also regarded the requirements under the section 'pace and rhythm' as very important.

Our result shows that online meetings allow for more and shorter workshops, which in turn can speed up the pace in the co-design process. The total workload during the whole design cycle does not need to be affected but will be distributed differently. We do not argue that an increased number of workshops are always better. We rather argue that online activities allow for a closer interaction within a team of people working with a design issue. Our experience align with blog posts from designers who have shared their experiences from online collaboration activities during the pandemic, discussing the importance of preparation, facilitation and finding a flow (Ringrose 2020), or raising the possibility of having more frequent but shorter workshops creating a new pace for design (Chao 2020). There might be an increased cost for additional staff to deliver the inclusive methods presented in the requirements. Our experience is that such costs can be balanced by less payments for travel, accommodation and for renting physical spaces.

Constantin et al. (2021) studied distributed participatory methods during COVID-19 among children and argued, that the most appropriate technology needed further investigation, and issues of access to the technologies needed to be addressed. Our result shows that online collaboration demands the use of a purposeful collection of digital tools. This aligns with blog posts during the pandemic who presented useful tools for online co-design activities (Chao 2020; Kennedy 2020). It seems that for many of those online tools, accessibility has not been in mind when developing the tool. Accessibility seems to have been bolted on afterwards. Nevertheless, we succeeded in finding online tools that were perceived accessible by most participants, at least after providing adapted instructions, teaching and support. In our case, it was the onboarding procedure when a participant started to use Zoom or Miro that demanded adapted and sometimes personal instructions. The need for support was most important in the beginning of the design process. Participants found it easier to collaborate when using a computer compared to using a smartphone. This is in line with findings by Slingerland et al. (2022). It seemed that those who always connecting to the internet by phone had developed strategies to cope with using the small screen. For example, they asked the meeting manager to zoom in on details.

Our results emphasise that successful online collaboration demands close attention to details and thorough preparation. This meticulous attention to detail is presented as scaffolding by Johansson (2019), who claims that it is rather the preparation of a participatory activity than the participatory method itself that needs to be adjusted when collaborating with users with impairments. This contradicts Dirks (2019) who claimed that there is no methodologically sound way to work in research teams where some members have cognitive impairments. Our findings are more in line with Kusunoki and Sarcevic (2012) who argued that the selected methods should be integrated and woven into the design process. We used a Miro-board as an online place where we could document the current version of the requirements. We believe this is important. When the participants are remote there needs to be something more than the online meeting to sustain engagement over time. Working towards a common goal and see the work progress is one way to withhold engagement over time. Another approach was used by Zhang et al. (2022) who used a gamified tool to promote engagement when doing online co-design. Both cases show that engagement can be enhanced by the use of digital tools. The need of rethinking collaboration when physical meetings is impossible or impractical is taken even further by Davis et al. (2021) who presents several forms of low contact and online collaboration where some also are asynchronous and spatiotemporal. They discuss how different settings can allow for a broadened participation from people with different sociocultural background. We believe that this kind of experimentation has a potential to offer enhanced opportunities for people with disabilities to participate in co-design. The democratic ambition in participatory design calls for new perspectives on participation (Björgvinsson, Ehn, and Hillgren 2010). It is an ongoing work to find ways for otherwise marginalised groups to be able to participate in and have influence on both the design process and the outcome of the design process. Accessible online collaboration in co-design activities seems to open a space and create the time for people with impairments to contribute with their perspectives and lived experiences.

People who participate in many online meetings can experience fatigue (Wiederhold 2020) and indeed, in our study participants also experienced fatigue sometimes but our experience is that they often found online meetings energy saving as opposed to physical meetings. Sometimes a participant needed to take a break and reconnect to the workshop at a later stage or leave a session and return at the next session. The overall impression though was that the participants experienced less fatigue at online workshops as compared to physical meetings and also could mitigate the consequences of fatigue more easily. By following the suggested requirements for online collaboration many of the potential downsides of an online meeting can be mitigated.

We had no dropouts in our study. However, engagement in PD might decline over time (Pilemalm and Timpka 2008). The people who participated in this study can be described as very committed. We can see some success criteria at play. The problems addressed in the study came from prioritising done by the participants community and members in that community wants to promote change towards a more inclusive society. And some of the participants was also engaged in planning of the projects. All participants were offered payment, but not all wanted to be paid. We believe that the strong engagement in the cause has a stronger explanatory power than the payment when trying to understand way people stayed in the projects. There was no extra payment for working with this guidelines. This work was done as an add-on to the projects as we realised that our experiences could help other less experienced to understand how to do collaborative work in online meetings.

Neither in-person workshops nor remote collaboration are in themselves novel concepts, although online co-design has become more frequent over time. What this study adds to already existing knowledge is how to include people with cognitive impairments in online co-design activities. The requirements presented are the ones prioritised by people with cognitive impairments and even if some requirements can be found in other recommendations it is the complete set of requirements that need attention to make a co-design session accessible. We regard this overall picture as our main contribution to the existing knowledge on how to include participants with diverse needs in online co-design.

5.1. Limitations

The focus for this study was to develop guidelines for accessible online collaboration for co-design activities when some participants have impairments. We did not focus on potential disadvantages with online collaboration.

The design process we used was targeted to create digital artefacts. We suggest that these requirements for online collaboration in the design process might be primarily applicable for co-design of digital artefacts. Online collaboration in the design process is probably more difficult if the aim for the design is a tangible object. Still, we believe that the requirements for online

collaboration are transferrable to many design processes.

It might be considered a limitation that the participants developing this guideline knew each other before they started with online collaboration. Would there be other requirements if participants don't know each other? Or would the path to the final result have been more complicated? Maybe, but we believe that strong emphasis on the following requirements could serve as onboarding tools for people who have never met before; Prepare guests and new participants for taking part; Present all participants with images and short bio; Create a soft and welcoming start to set a convivial tone. We also suggest that the agenda of the first workshop need to have activities where participants present themselves and do activities to learn to know each other serves to overcome difficulties understanding each other and empathising with each other when they have never met in person before.

The participants where highly motivated to take part in the study. Recruiting a group of participants being less motivated could have prolonged the process of arriving to the requirements and it could also be a higher risk of participant dropouts. This emphasises the importance to collaborate with participants already when prioritising the research topic (Bødker and Kyng 2018). A topic that aligns with what is considered important to the people who chose to opt-in and participate decreases the risk of participant drop out. To be able to participate online could even be regarded as a measure to provide for participants to stay in the process.

6. Conclusion

Online collaboration for user participation in co-design provides some clear benefits compared with on-site collaboration. The most prominent is that participating from online is a way to include people with impairments in the design process. The participants do not need to travel, and it is possible to stay in a familiar environment, avoid stressful moments and save energy. Reducing travelling is also a more sustainable way of collaborating. We suggest that a co-design process can be strengthened by transition to online collaboration, owing to the possibility of having more frequent interaction, allowing for an effective design process since real-time design changes can be undertaken instantly, saving the participants' energy, and enabling the traditionally non-transparent parts of the design process to be more transparent. Using accessible online tools, careful planning and preparation of the meeting and establishing a trustful and convivial working atmosphere during each activity are the cornerstones for accessible online collaboration. People with impairments can participate in online collaboration if the activities are scaffolded and arranged to meet accessibility requirements. The requirements for online collaboration proposed in this study creates a foundation for equal and democratic collaboration and participation in the design process, especially for people with impairments. Online participation can for some people with impairments be the most accessible way or even the only way to participate in a design process. Our proposed requirements present important prerequisites for how to scaffold the online activities to be accessible.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by Forskningsrådet om Hälsa, Arbetsliv och Välfärd.

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